Warning: This task has a 10-minute time limit for submitting your solution. Once you request an input, the timer will start automatically, and after the deadline it will no longer be possible to submit a solution for that input. *You can always request a new input*, so don't worry if the timer expires: you will simply need to request and download a new input.

To help you with this task, we have prepared some ***solution templates****, which only include the parts for reading the input and writing the output (from the keyboard and to the screen). You can decide whether to read/write to files by uncommenting the appropriate lines of code.*

- *Download the template in C: gossip.c*
- *Download the template in C++: gossip.cpp*
- *Download the template in C#: gossip.cs*
- *Download the template in Go: gossip.go*
- *Download the template in JavaScript: gossip.html*
- *Download the template in Java: gossip.java*
- *Download the template in Pascal: gossip.pas*
- *Download the template in Python: gossip.py*
- *Download the template in VisualBasic: gossip.vb*

Problem Description

Rumors are already starting to spread among the participants about who will be the next winner of the *Italian Olympiad in Informatics*.



Among the N participants of the OII, numbered from 0 to $N-1$, there are M friendships, the i-th of which establishes that people A_i and B_i are friends. The friendship network is connected: for every pair of participants, there is at least one chain of friendships connecting them.

On the first day, K gossips are born, numbered from 0 to $K-1$: the i-th of which is started by person G_i, who believes that gossip right from the start.

In the following days, anyone who believes a gossip shares it with their friends. During each day, the gossips are shared in index order: first gossip 0, then 1, and so on up to $K-1$.

The participants are very predictable: they only believe the first gossip they hear. Once convinced of a gossip, they will share it starting from the next day and will ignore all the others.

After a certain number of days, every participant will believe a gossip and the rumors will stop spreading.

Determine for each of the N participants which gossip they will believe once the rumors have completely spread.

> **Warning**: To avoid large output files, you will need to print the result as follows.
>
> Let P_i be the gossip that the i-th participant will eventually believe,

print

$$\sum_{i=0}^{N-1} P_i \cdot (i+1)$$

which is the sum of the final gossips, each multiplied by the participant's index increased by 1.

Warning: This quantity is too large to be represented with a 32-bit integer, remember to use an integer type of at least 64 bits.

Input Format

The first line of the input file contains an integer T, the number of test cases. T test cases follow, numbered from 1 to T. Each test case is preceded by an empty line.

Each test case is composed as follows:

- a line containing the three integers N, M, K.
- a line containing the K integers $G_0, \ldots, G_{K-1}$.
- M lines, the i-th containing the two integers A_i, B_i.

Output Format

The output file must contain the answer to the test cases that you managed to solve. For each test case you solved, the output file must contain a line with the string "`Case #test:`", where `test` is the test case number (starting from 1), followed by the 64-bit integer `ris`.

Constraints

- $T = 20$, in the input files you will download there will be exactly 20 test cases.
- $1 \leq K \leq N \leq 10^5$.
- $1 \leq M \leq 10^5$.
- $0 \leq G_i \leq N-1$, for each i such that $0 \leq i \leq K-1$.
- $0 \leq A_i, B_i \leq N-1$, for each i such that $0 \leq i \leq M-1$.
- The pairs of friendships are all distinct.
- There are no people who are friends with themselves.
- The friendships form a connected graph.

Testcase Groups:

The test cases are divided into 10 groups.
Each group is worth 5 points which are obtained if all the test cases of the group are solved correctly. Different groups may contain test cases satisfying the same constraints.

Constraints	Groups
$K = 2, A_i = i, B_i = i + 1,$ for each $0 \leq i \leq N - 1$	$1, 2$
$A_i = i, B_i = i + 1,$ for each $0 \leq i \leq N - 1$	$3, 4, 5$
$N \leq 5000, M \leq 5000$	$6, 7, 8$
No additional constraints	$9, 10$

Input/Output Examples

Input:

```
4

5 4 3
4 1 2
0 1
0 2
0 3
0 4

5 6 2
4 1
0 2
0 3
3 4
3 1
1 2
1 4

5 4 2
1 3
0 1
1 2
2 3
3 4

7 6 3
3 1 6
0 1
1 2
2 3
3 4
4 5
5 6
```

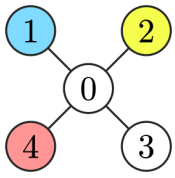
Output:

```
Case #1: 8
Case #2: 5
Case #3: 9
Case #4: 29
```

Explanation

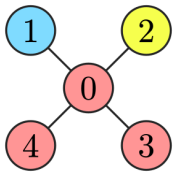
In the **first example case**, the gossip graph is initially as follows:

The gossips are shared in the following way:



- After one day, participant 0 starts believing gossip 0, shared by participant 4.
- After two days, participant 3 also starts believing gossip 0.

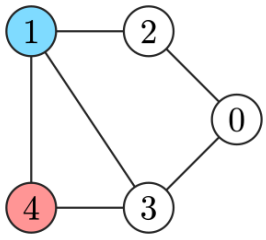
The final graph is therefore as follows:



Thus, the participants will respectively believe the gossips $[0, 1, 2, 0, 0]$. The integer to print is thus $0 \cdot 1 + 1 \cdot 2 + 2 \cdot 3 + 0 \cdot 4 + 0 \cdot 5 = 8$.

In the **second example case**, the graph is initially as follows:

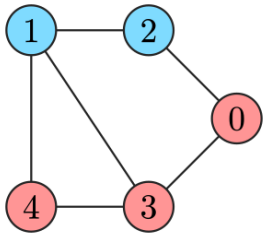
The gossips are shared in the following way:



- After one day, participant 3 starts believing gossip 0, and participant 2 starts believing gossip 1.

- After two days, participant 0 starts believing gossip 0.

The final graph is therefore as follows:



Thus, the participants will respectively believe the gossips $[0, 1, 1, 0, 0]$. The integer to print is thus $0 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 + 0 \cdot 4 + 0 \cdot 5 = 5$.

In the **third example case**, when no gossip is spreading anymore, the participants will respectively believe the gossips $[0, 0, 0, 1, 1]$. The requested integer is therefore $0 \cdot 1 + 0 \cdot 2 + 0 \cdot 3 + 1 \cdot 4 + 1 \cdot 5 = 9$.

In the **fourth example case**, when no gossip is spreading anymore, the participants will respectively believe the gossips $[1, 1, 0, 0, 0, 2, 2]$. The requested integer is therefore $1 \cdot 1 + 1 \cdot 2 + 0 \cdot 3 + 0 \cdot 4 + 0 \cdot 5 + 2 \cdot 6 + 2 \cdot 7 = 29$.